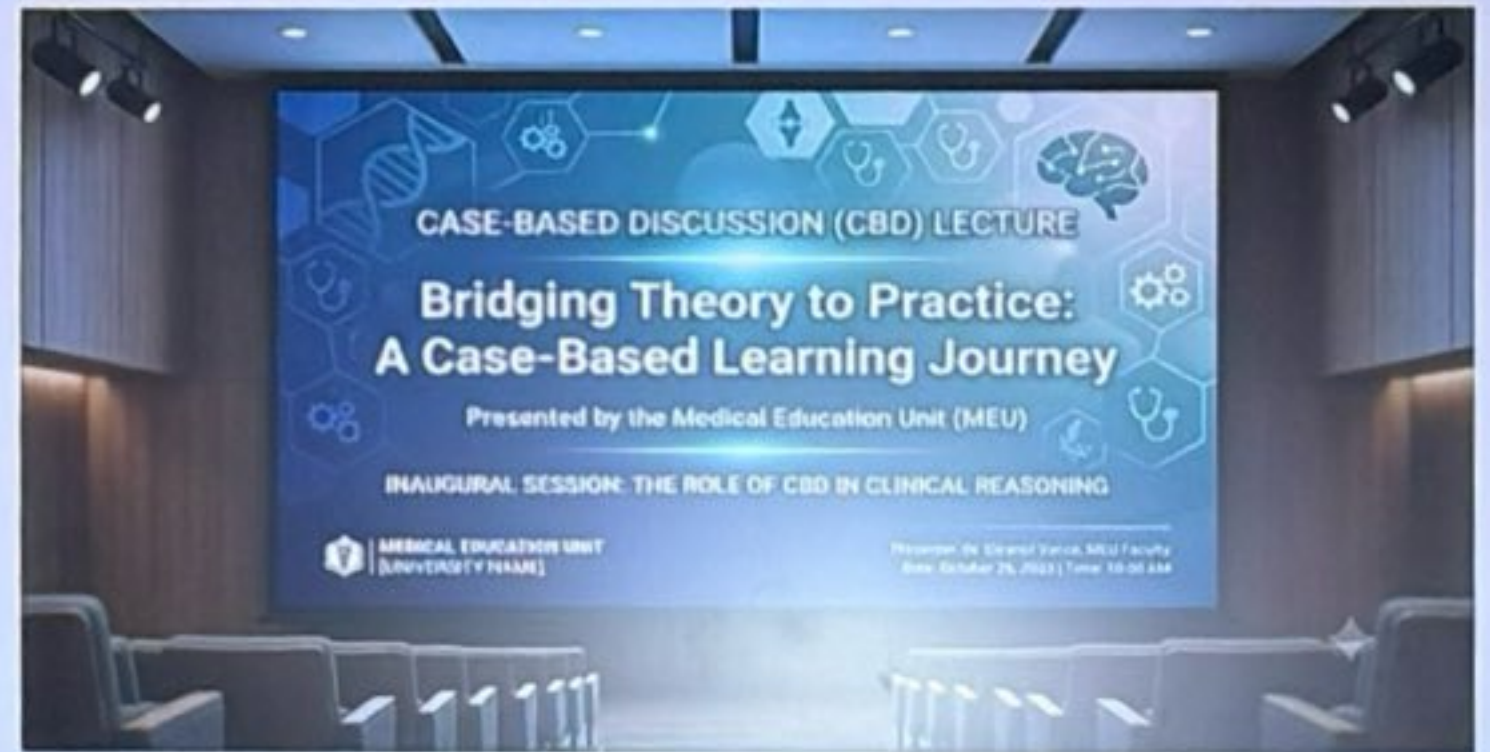




Al-Ameed College of Medicine

Medical Education Unit

CBD-TBL



3rd Year

Academic Year : 2026-2027

CBD-TBL Session (UGY3)

The Crushing Chest Pain: When Every Minute Counts



second stage

CBD-TBL | Acute Coronary Syndrome

Medical EDU Unite

➤ **Learning Objectives**

- By the end of this session, students should be able to:
- Explain the pathophysiology of acute coronary syndrome (ACS)
- Differentiate STEMI, NSTEMI, and unstable angina
- Understand pharmacologic management of ACS
- Apply acute reperfusion algorithms in STEMI
- Integrate health promotion and secondary prevention in coronary artery disease

Case Presentation

Mr. Ali is a 55-year-old man who presents to the emergency department **with severe central chest pain that started 1 hour ago while climbing stairs.**

He describes the pain as a **crushing**, heavy pressure **radiating to his left arm** and jaw, associated with sweating, nausea, and shortness of breath. **The pain has not resolved with rest.**

He has a history of type 2 diabetes mellitus and hypertension, both poorly controlled. He is a current smoker (20 pack-years) and has a family history of ischemic heart disease (father had an MI at age 50).

He has never experienced chest pain like this before.

Mr. Ali asks:

"Is this just indigestion... or is my heart in danger?"



Social & Lifestyle History

Current smoker (20 pack-years)

Sedentary lifestyle, high-fat diet

Poor adherence to antihypertensive and antidiabetic medications

High occupational stress

No alcohol or illicit drug use

Physical Examination (During Episode in ED)

Vital Signs:

HR: 105 bpm (regular)

BP: 150/95 mmHg

RR: 24/min

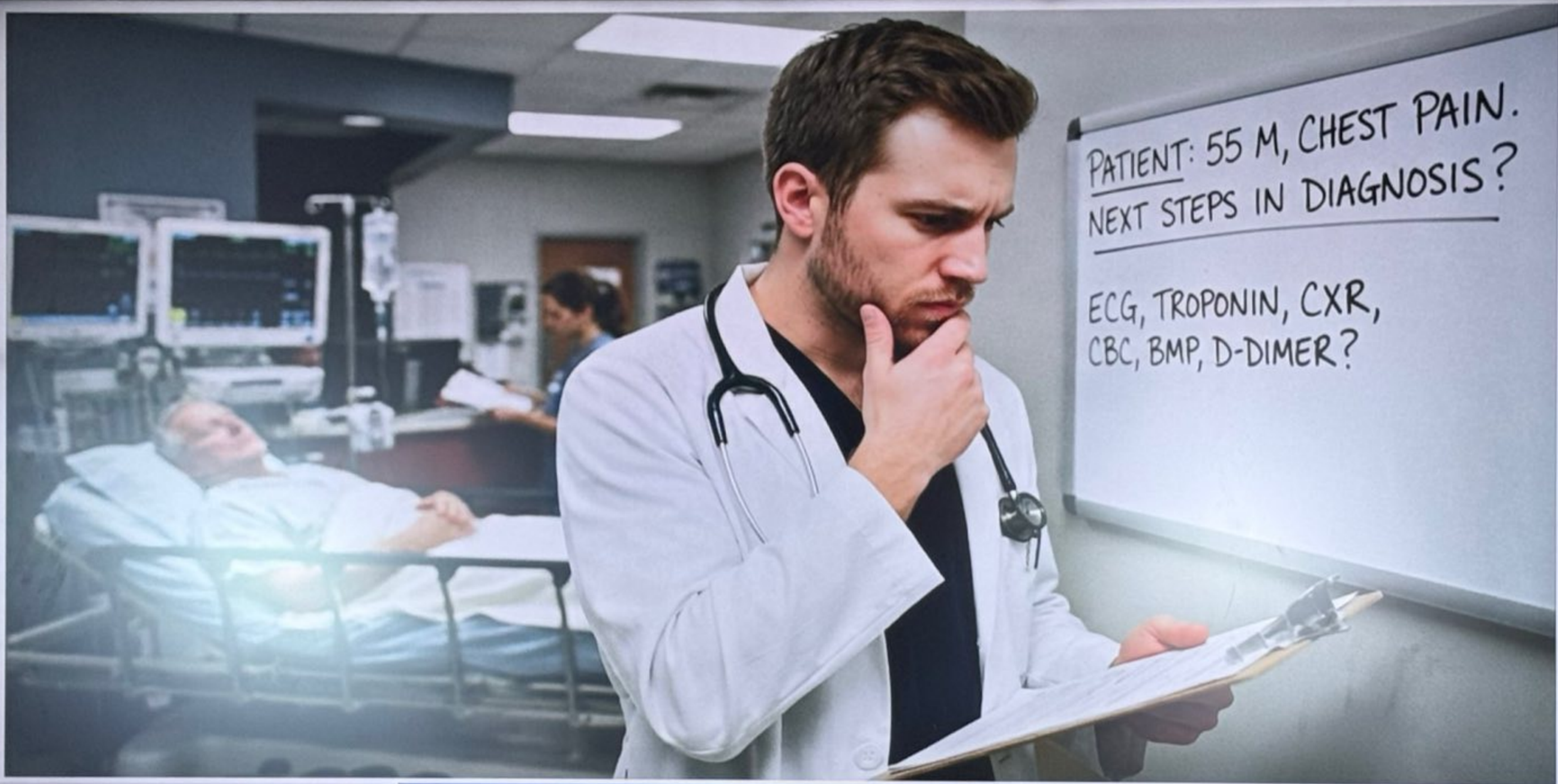
O2 sat: 94% RA

Cardiovascular EXAMINATION :

Diaphoretic, anxious-looking

S4 gallop present

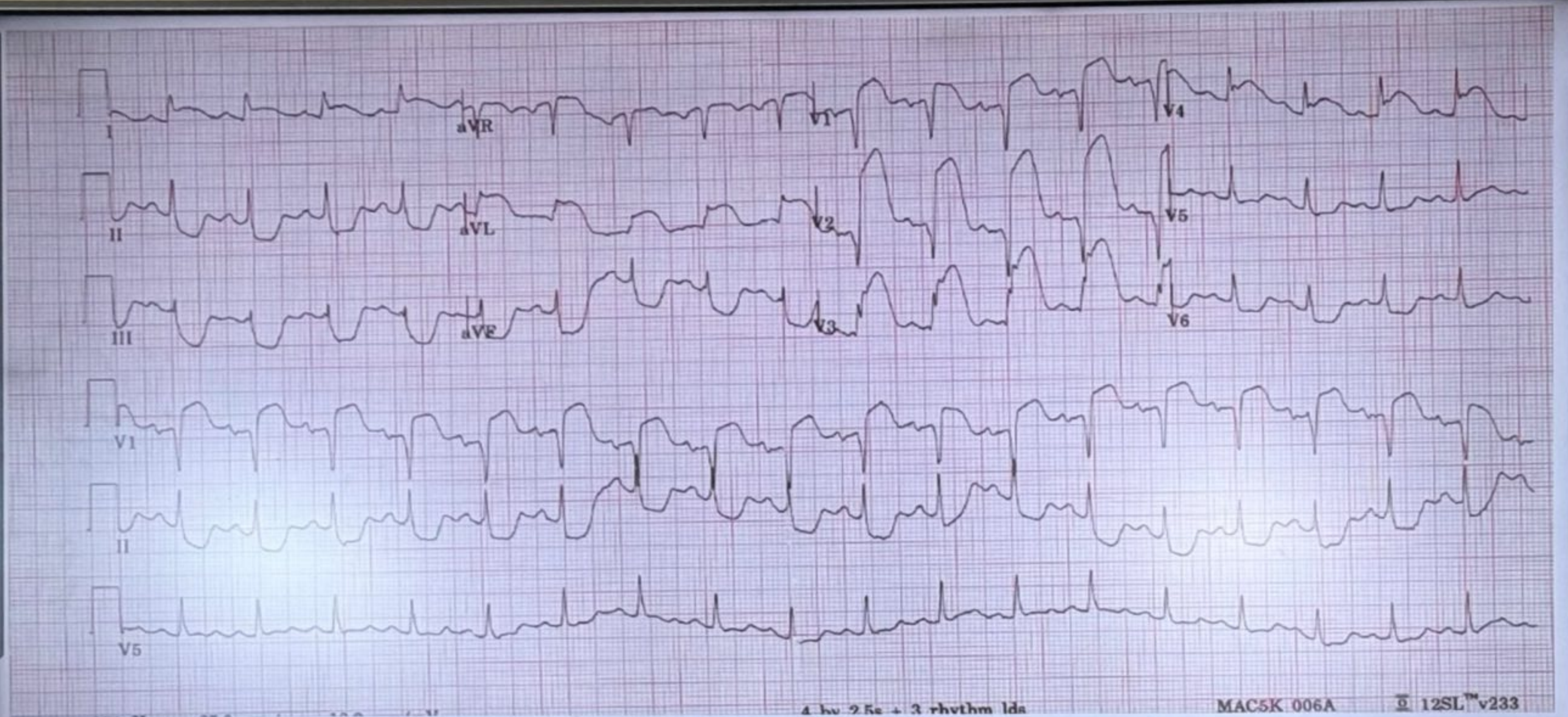
No murmurs



second stage

CBD-TBL | Acute Coronary Syndrome

Medical EDU Unite



ECG: ST-segment elevation > 2 mm in leads V2-V4-AVL-L1 (anterior and lateral lead)

Troponin I: markedly elevated

second stage

CBD-TBL | Acute Coronary Syndrome

Medical EDU Unite

task1-What is the differential diagnosis of chest pain in internal medicine?

Task2-what is the Pathophysiology of ACS?

Task 2 – Mechanism Mapping?

Task4-Which features suggest ACS rather than musculoskeletal or non-cardiac chest pain

Task5-Aspirin, in low doses (75-325mg), acts as antithrombotic. Explain why does it act as antithrombotic at such low doses only?

Task6-Metoprolol is a β_1 -selective (cardioselective) beta blocker; how does it differ from other cardioselective beta blockers such as atenolol?

Task7-Why Early Reperfusion Works?

Task 8 – Biopsychosocial Analysis

Identify the factors contributing to his condition using the Biopsychosocial Model.

1. The primary mechanism of acute coronary syndrome is:

- A. Coronary artery vasospasm alone
- B. Atherosclerotic plaque rupture with thrombosis
- C. Viral myocarditis
- D. Valvular prolapse

2. Aspirin reduces platelet aggregation by:

- A. Blocking β_1 receptors
- B. Irreversibly inhibiting COX-1
- C. Blocking ADP receptors
- D. Inhibiting thrombin

3. A red flag feature in a patient with chest pain is:

- Pain reproducible with palpation
- Pain relieved by antacids
- Radiation to the jaw with diaphoresis
- Sharp pain worsened by breathing